

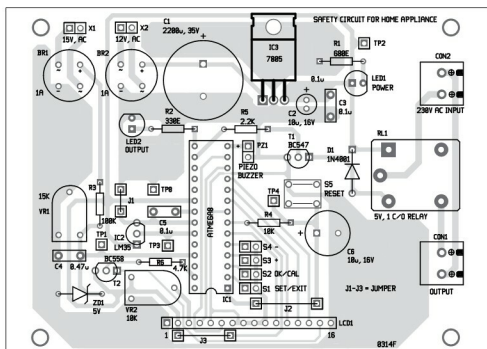


Fig. 6: Component layout for the PCB

TABLE III
Software Calibration Procedure

Steps	Descriptions
Step 1	• Plug your unit and the multimeter into the stabiliser's output.
Step 2	• Press the OK/CAL button for about 2 seconds and observe the calibration menu. Maine AC voltage calibration Mode (Anytime you want to exit the calibration routine, press SET/EXIT button, but it is not recommended to do so until you complete it.)
Step 3	• The menu will ask to decrease your stabiliser's voltage. Now decrease the input AC voltage • Wait for the next screen (it will pop up automatically after 3 seconds).
Step 4	• Now, first it is recommended to set the stabiliser voltage between 70V and 130V. • Observe the following screen now asking you to set the low point. Set low points 100 • Whatever you set your stabiliser's output voltage (also displayed on your multimeter), enter it here by using the '+' and '-' switches. • After setting, press OK/CAL button again.
Step 5	• Now the menu will ask you to increase the stabiliser's voltage. Now increase the input AC voltage • Wait for the next screen (it will pop up automatically after 3 seconds).
Step 6	• Now it is recommended to set the stabiliser's voltage between 280V and 330V. • Observe the following screen now asking you to set the high point. Set high points 300 • Whatever you set your stabiliser's output voltage (also displayed on your multimeter), enter it here by using the '+' and '-' switches. • After setting, press OK/CAL switch again.
Step 7	• The device will then show the calculated calibration constants and amend them automatically to its internal sampling routines and will start following them. • 'Calibration Successful' message at the end will depict calibration success and the display will revert back to main screen. You are done now.

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com website.

sembly errors. Use IC base for micro-controller IC1. The switches should be mounted on a firm base and connected to the PCB through wires. To test the circuit for proper functioning, verify voltage levels as indicated in test point Table II.

Calibration procedure

For accurate mains voltage measurements and cut-off functioning, the device must be calibrated with respect to the mains voltage. The calibration can be done through hardware or software. The hardware method employs voltage calibration by VR1, while the software method uses the internal software calibration routines with a variable external AC voltage generator or auto-transformer.

For the first time, set VR1 to 10k before powering the unit. Now switch on the unit and observe the mains voltage on the LCD. Connect your multimeter to the mains socket. Now trim VR1 to match the unit's LCD reading to the multimeter's reading, and you are ready to go.

The software method can additionally be used to enhance the accuracy. For software calibration, you will require an external stabiliser or auto-transformer that is capable of providing AC voltages from 100V to 300V or more. Follow the directions shown in Table III and your system is ready to use. ●

The author is a B.Tech. in electronics and instrumentation



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